HEAR: State-Constrained LLM Feedback for Museum Sustainability

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Abstract—Museum environmental control protects collections but can increase energy use. This study developed Heritage-Ecology Adaptive Reflection (HEAR), a browser-based museum game that asks visitors to balance collection preservation, environmental protection, and energy load. A deterministic engine updates three state values, while a Large Language Model (LLM) may select one reflective prompt from a fixed candidate set at preset checkpoints; it cannot change scores or decide for the player. A 30-participant pilot compared HEAR with a control version using the six-item Heritage-Ecology Trade-off Understanding (HETU) scale and exploratory mechanism measures. HETU reliability was high at pre-test and post-test, but the baseline-adjusted HEAR effect was not detected (p = .832). Mechanism evidence differed: HEAR participants had a higher Trade-off Reasoning Score than controls (1.20 vs. 0.07, U = 190.5, p < .001), and explicit multi-factor reasoning appeared in 40% of HEAR responses and 0% of control responses. HEAR also received high ratings for reflective engagement, feedback utility/congruence, agency preservation, and interaction non-interference. The findings suggest immediate reflective benefits.

Keywords—Museum environmental control, heritage-ecology, human-computer interaction, large language models, reflective engagement

I. Introduction

Museum environmental control concerns collection preservation, visitor comfort, and environmental cost. Recent museum-environment research shows that fixed temperature and humidity targets can increase energy use and may not fit every collection or building context [1]. The practical question is how museums should balance collection care against energy use. Conservation literature has begun to address this issue directly. Debates over relative humidity and temperature standards emphasize risk-based parameter setting rather than a single fixed indoor environment [2], [3]. Recent adaptive climate-control work also links collection monitoring with sustainability outcomes [4]. These studies acknowledge the trade-off at an institutional level. Visitors, however, rarely make such decisions themselves while viewing objects in museums and galleries.

The interactive heritage project developed in this study turns an abstract policy problem into a concrete experience in which visitors weigh consequences and make decisions. Tangible interfaces have been used to improve learning in virtual museums [5]. Immersive experience is also associated with perceived usefulness in virtual reality exhibitions [6]. Broader XR heritage research likewise treats immersive technology as a way to connect audiences with cultural settings and consequences [7]. These interfaces make outcomes visible, but they do not directly test whether visitors consider collection condition and external environmental burden together.

Large Language Models (LLMs) can provide context-sensitive educational feedback [8], although fluent output may still contain unsupported claims [9]. Prompt design can restrict input [10]. Retrieval can anchor responses in external evidence [11], and self-checking can expose inconsistencies [12]. Agent systems can connect LLMs to actions [13] or simulate behavior over time [14]. Critical research shows why fluent language should not be treated as grounded understanding [15]. Museum learning systems therefore need a clear boundary between measured state, model language, and player action.

Heritage-Ecology Adaptive Reflection (HEAR) builds this boundary into the game. At fixed decision points, the browser sends only a verified state and approved candidate set. The model returns an index, not an action or score. Control participants follow the same rules without model-selected prompts. The primary analysis tests HETU change after adjustment for background and baseline score. Exploratory analyses examine open responses and HEAR ratings. Because these analyses were specified after questionnaire review, they provide evidence of mechanisms, not confirmatory evidence of learning.

II. Methodology

A. Study Design, Data Sources, and Participants

Data were collected on 25 August 2026 through one pre-test and two post-tests tailored to the HEAR and control conditions. The study design listed 30 participant IDs, with 15 in each condition. Disciplinary background was classified as art-related or non-art-related. The four cells were art-related HEAR (n = 8), art-related control (n = 7), non-art-related HEAR (n = 7), and non-art-related control (n = 8). Table 1 gives the cell counts. The available records did not describe the allocation procedure. The study is therefore reported as a user trial, not a randomized trial.

The pre-test received 31 submissions. Four data-entry errors were corrected by comparing the preset IDs with post-test records, and one duplicate submission was removed. After cleaning, each of the 30 planned IDs had one pre-test and one post-test. All 30 participants reported reaching a final ending.

TABLE I. SAMPLE BY CONDITION AND DISCIPLINARY BACKGROUND

Background	Control	HEAR	Total
Art-related	7	8	15
Non-art-related	8	7	15
Total	15	15	30

B. HEAR Architecture and Game Flow

HEAR separates interaction, language, and analysis into three layers, as shown in Figure 1. The browser converts each player action into changes in collection preservation (M), environmental protection (E), and energy load (N). The feedback module reads the verified state and selects one reflective prompt from an approved set. The analysis layer links questionnaires to participant IDs. When event records exist, the same IDs can connect questionnaires with game events. Score calculation remains reproducible if the cloud model changes or is temporarily unavailable.

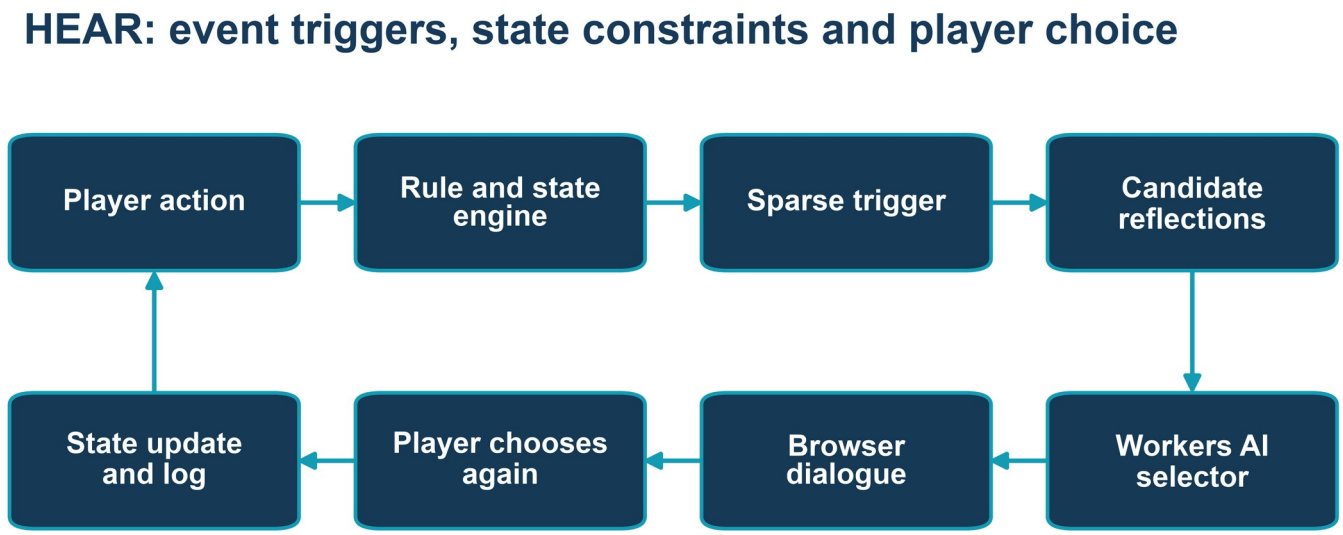


Fig. 1. State-constrained HEAR architecture

1) Deterministic Interaction Cycle

The game has four stages, as summarized in Table 2 and Figure 2. In Stage 1, the player cleans the venue and a painting, then decides at 50% completion whether to continue. Stage 2 combines temperature, humidity, and air-purification settings with an environmental calibration task controlled by the WASD keys. In the darkened gallery of Stage 3, the player chooses no light, one light, or three lights. Stage 4 presents the final state before the player maintains or stops the system. These choices lead to the distinct outcomes Ending A and Ending B. Both conditions use the same interface, options, and state rules. Only the HEAR group receives reflective prompts at preset checkpoints.

$$x_{t+1} = clip\left(x_t + \Delta a_t, 0, 100\right), \quad x_t = \left(M_t, E_t, N_t\right)^T \tag{1}$$

Equation (1) defines the deterministic state update. In this equation, $x_t = (M_t, E_t, N_t)^T$ is the state vector at decision step t, and x_{t+1} is the updated state. M_t, E_t, and N_t denote collection preservation, environmental protection, and energy load, respectively. The action a_t contributes the fixed change vector Δa_t. The superscript T denotes transposition. The clip function acts on each component and restricts it to [0, 100]. Model-generated text does not enter this calculation.

TABLE II. FOUR-STAGE INTERACTION AND FEEDBACK TIMING

Stage	Player action	Decision point
1	Clean mist and dust from the painting	Stop or continue at 50% cleaning
2	Adjust environmental factors and stabilize the calibration sphere	Keep or change the current settings
3	Choose no light, one light, or three lights	Select a lighting plan
4	View the final state	Maintain or stop the system

2) State-Constrained LLM Feedback

At each feedback trigger, the browser serializes the current stage, M/E/N values, selected option, and available actions. It also loads the candidate set written for that checkpoint. The server must return one integer index. The browser accepts the value only if it belongs to the candidate set. Otherwise, it displays a state-matched fallback prompt. Table 3 summarizes this boundary and the reserved audit fields. The design adopts the evidence-constraining logic of retrieval systems [11] without giving the model control of the interface.

$$j^* = arg\ max_{j \in \{0,\ldots,K-1\}}\ P_\theta\left(j \mid F_t,\ C_t\right) \tag{2}$$

Equation (2) expresses the constrained selection policy. At decision step t, F_t is the verified state, C_t is the approved candidate set, j indexes a candidate, K is the number of candidates, and θ denotes the model parameters. $P_\theta(j \mid F_t, C_t)$ is the model-assigned selection probability for candidate j. The arg max operator returns j^*, the index with the highest value. Because j^* must lie in {0, ..., K - 1}, the model can

select only an approved prompt. The player still controls the next action.

TABLE III. HEAR FEEDBACK BOUNDARY AND AUDIT FIELDS

Step	System action	Audit field
1	Check whether feedback is allowed at the current checkpoint	participant, sessionId, condition, checkpointId, stage
2	Build the verified state and approved candidate set	response, verifiedFact
3	Request and validate one candidate index	feedback_source, model_name, request_success
4	Display the selected candidate or a matched fallback prompt	fallback_used, latency_ms
5	Record the event after feedback, then return the next choice to the player	recordedAt and subsequent action



Fig. 2. Four game stages and the Ending A and Ending B outcomes

C. *Measures and Evidence Boundaries*

The primary outcome was HETU, calculated as the mean of six pre-test items and six post-test items rated from 1 to 7. The items covered energy and resource use, delayed external burdens, short-term and long-term trade-offs, context-dependent climate standards, alternatives to binary choices, and multi-criteria decisions. Higher scores indicate stronger recognition of trade-offs between heritage preservation and ecological goals. The overall mean used only the six matched pre-test and post-test items. The pre-test prior-knowledge item was excluded.

Both post-tests asked which choice in the game caused the most hesitation and why. Open responses were scored from 0 to 3, unlike the 1-to-7 post-test items. A score of 0 indicated no assessable trade-off reasoning or only an operational difficulty. A score of 1 identified one goal or consequence. A score of 2 required an explicit comparison of at least two goals or consequences. A score of 3 also addressed a long-term or system-level trade-off. Ambiguous responses received the lower score. One coder rated the de-identified responses, so this outcome is reported as exploratory between-condition evidence.

The HEAR post-test also covered numeric comparison, delayed cost, reconsideration, operational clarity, feedback-state congruence, non-interference, player agency, trade-off support, and the next choice. The Reflective Engagement Score (RES) averages the reconsideration and feedback-prompted change items. The Perceived Feedback Utility/Congruence score (PFUC) averages feedback-state congruence and trade-off support. For each participant:

$$RES_i = \frac{Q_{R,i} + Q_{FC,i}}{2} \quad (3)$$

$$PFUC_i = \frac{Q_{C,i} + Q_{TS,i}}{2} \quad (4)$$

In Equations (3) and (4), i indexes participants. RES_i and $PFUC_i$ are participant-level arithmetic means. $Q_{R,i}$ denotes reconsideration, $Q_{FC,i}$ denotes feedback-prompted change or reconsideration, $Q_{C,i}$ denotes feedback-state congruence, and $Q_{TS,i}$ denotes trade-off support. All four items used a 1-to-7 scale. The two post-tests contained three broadly similar concepts, but their wording differed. The corresponding between-condition comparisons are therefore exploratory rather than formal tests of identical measures.

D. Statistical Analysis

Cronbach's alpha estimated scale consistency. Two-sided paired t tests compared pre-test and post-test HETU within each condition, with Wilcoxon signed-rank tests as sensitivity analyses. The primary between-condition estimate was HEAR change minus control change, tested with Welch's t test and reported with a 95% confidence interval and Hedges' g. A two-factor change-score analysis of variance tested condition, disciplinary background, and their interaction.

TABLE IV. MEASURES AND ANALYSIS PLAN

Research question	Outcome	Analysis
Did each condition change?	Post-test minus pre-test HETU	Paired t test and Wilcoxon sensitivity analysis
Was change greater in HEAR?	Difference in HETU change	Welch t test, 95% CI, and Hedges' g
Did background modify change?	HETU change	Two-factor analysis of variance with interaction
Was the result robust to baseline control?	Post-test HETU	HC3 analysis of covariance with baseline and interaction
Which HETU items differed?	Six item-level change scores	Welch tests with Holm adjustment
How did participants rate the experience?	Condition-specific post-test items	Descriptive statistics and exploratory shared-item tests
Did participants articulate trade-offs?	Open-response score (0 to 3)	Mann–Whitney U, rank-biserial effect size, and Fisher sensitivity test
Did the HEAR mechanism operate as designed?	RES, PFUC, agency, and non-interference	HEAR descriptives, bootstrap 95% CIs, and Spearman–Brown reliability

Because baseline HETU differed numerically, a baseline-adjusted analysis of covariance used post-test HETU as the outcome and included baseline HETU, condition, background, and their interaction with HC3 robust standard errors. The exploratory Trade-off Reasoning Score was tested with a two-sided Mann-Whitney U test and a rank-biserial effect size. Fisher's exact test examined the proportion scoring at least 2. RES and PFUC were summarized with percentile bootstrap confidence intervals based on 50,000 resamples, and Spearman-Brown reliability was reported. The HEAR-only composites were not tested against the scale midpoint. Holm adjustment covered six HETU item tests and three approximately shared post-test measures. Analyses used Python 3.12.13, SciPy 1.18.1, and statsmodels 0.14.6. Given the pilot sample size, estimates and confidence intervals were emphasized over factor analysis. Table 4 summarizes the analysis plan.

III. RESULTS

A. Sample Completeness and Scale Reliability

Data cleaning produced 30 complete pre-test and post-test pairs, with no missing planned IDs or additional IDs. All 15 participants in each condition reported reaching the final page. HETU had high internal consistency at pre-test (alpha = .905) and post-test (alpha = .885). Post-test alpha was .903 in HEAR and .851 in the control group. Baseline HETU was 5.46 (SD = 1.12) in HEAR and 4.64 (SD = 1.15) in the control group. The difference was 0.81 (95% CI, -0.04 to 1.66, Welch p = .060, Hedges' g = 0.70). Although the confidence interval crossed zero, the difference indicated a higher starting point and less room for improvement in HEAR. Baseline therefore required control in the primary analysis.

B. HETU Change Within and Between Conditions

Figure 3 shows the pre-test and post-test trajectories. Control HETU rose from 4.64 (SD = 1.15) to 5.47 (SD = 0.77). Mean change was 0.82 (95% CI, 0.38 to 1.26), t (14) = 4.02, p = .001, Cohen's dz = 1.04. HEAR HETU rose from 5.46 (SD = 1.12) to 5.84 (SD = 0.95). Mean change was 0.39 (95% CI, -0.14 to 0.92), t (14) = 1.57, p = .139, dz = 0.40. Wilcoxon tests gave p = .004 for control and p = .116 for HEAR.

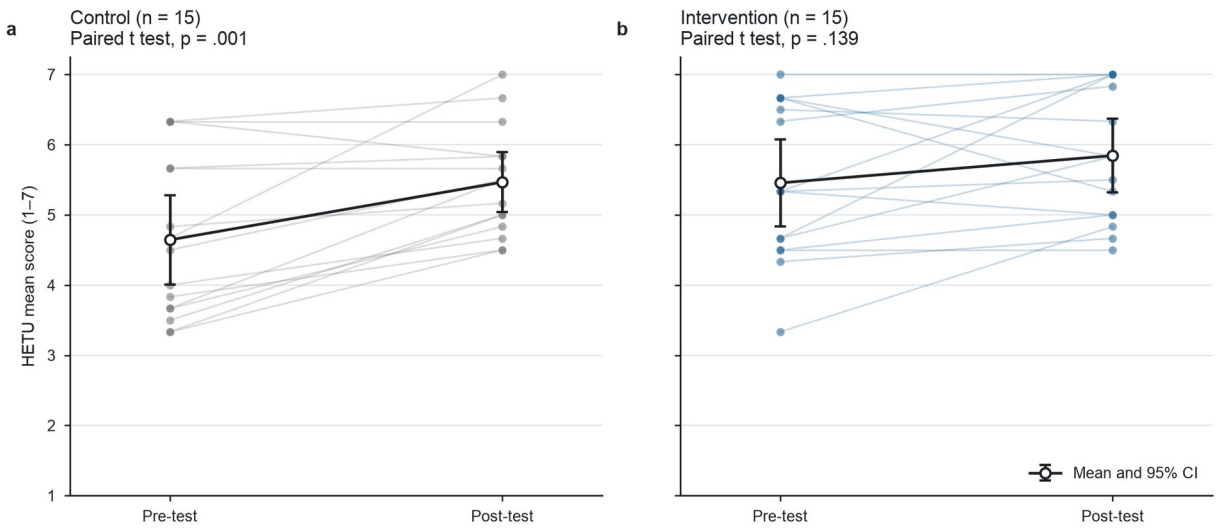


Fig. 3. Participant HETU trajectories by condition

The pilot detected no additional HETU change in HEAR relative to control. Mean change in HEAR was 0.43 points smaller than in control (95% CI, -1.09 to 0.23), Welch t(27.03) = -1.35, p = .189, Hedges' g = -0.48. A Mann–Whitney test gave p = .164. In the change-score analysis of variance, the condition effect was F(1,26) = 0.39, p = .540, partial eta squared = .015. The background effect was also F(1,26) = 0.39, p = .540. The condition-by-background interaction was F(1,26) = 0.16, p = .697, partial eta squared = .006. Figure 4 shows the overall and subgroup estimates.

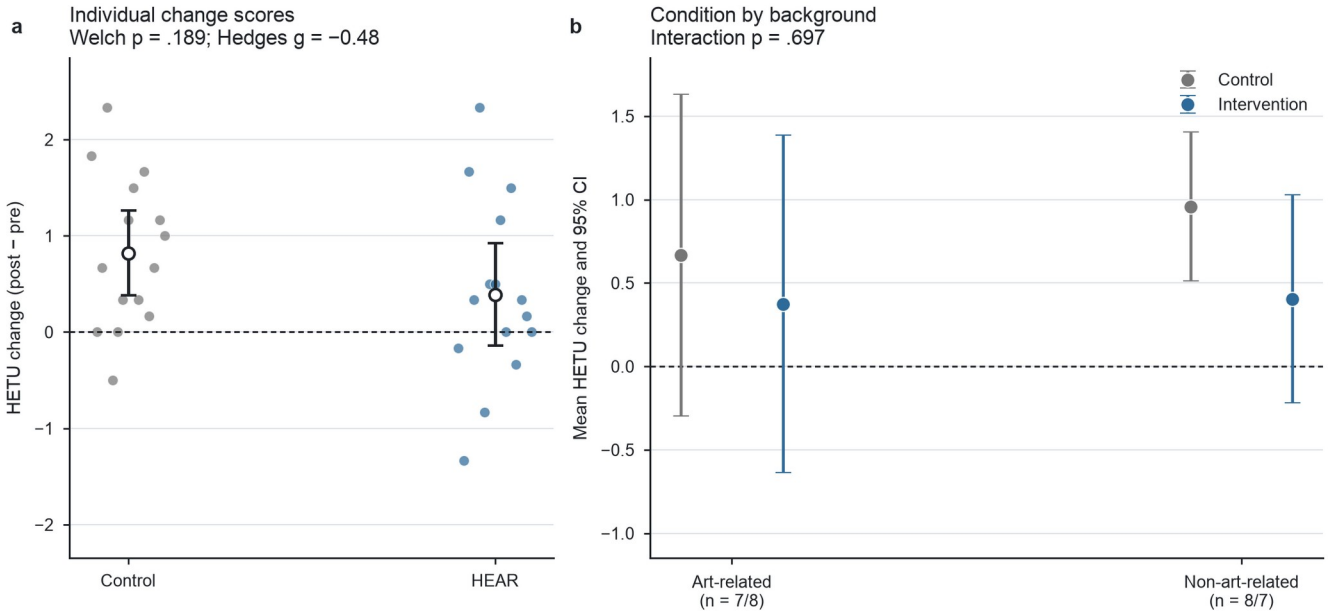


Fig. 4. Overall HETU change and disciplinary-background subgroup results

Baseline adjustment did not change the conclusion. In the HC3 analysis of covariance, the HEAR coefficient was -0.10 (95% CI, -1.02 to 0.82), p = .832. The condition-by-background coefficient was 0.21 (95% CI, -0.86 to 1.29), p = .695. Baseline HETU predicted post-test HETU (coefficient = 0.43, p < .001), identifying starting level as an important factor in this pilot. Tables 5 and 6 summarize the main HETU results and the combined mechanism evidence.

TABLE V. PRIMARY HETU RESULTS

Outcome	Control	HEAR	HEAR minus control	Inference
Pre-test HETU	4.64 (1.15)	5.46 (1.12)	0.81	95% CI -0.04 to 1.66, p = .060
Post-test HETU	5.47 (0.77)	5.84 (0.95)	0.38	Descriptive result
HETU change	0.82 (0.79)	0.39 (0.96)	-0.43	95% CI -1.09 to 0.23, p = .189, g = -0.48
Paired change	t(14) = 4.02	t(14) = 1.57		p = .001 and p = .139, respectively
Baseline-adjusted condition effect	Control (reference)	-0.10	-0.10	HC3 95% CI -1.02 to 0.82, p = .832
Condition by disciplinary background			0.21	HC3 95% CI -0.86 to 1.29, p = .695

TABLE VI. OUTCOME AND MECHANISM MEASURES

Measure	Results
HETU gain	Control: 0.82 (0.79); HEAR: 0.39 (0.96). HEAR minus control -0.43, 95% CI -1.09 to 0.23, p = .189, g = -0.48
Trade-off Reasoning Score (0 to 3)	Control: 0.07 (0.26), median 0 [0 to 0]; HEAR: 1.20 (0.94), median 1 [0.5 to 2.0]. U = 190.5, p < .001, rank-biserial r = .69
Explicit multi-factor reasoning	Control: 0/15 (0%); HEAR: 6/15 (40%). Fisher p = .017
RES (1 to 7)	Control: Not comparable; HEAR: 5.77 (1.15). 95% bootstrap CI 5.20 to 6.30, Spearman–Brown = .75
PFUC (1 to 7)	Control: Not comparable; HEAR: 6.03 (0.93). 95% bootstrap CI 5.57 to 6.47, Spearman–Brown = .82
Agency preservation (1 to 7)	Control: 5.13 (1.06)*; HEAR: 5.93 (1.10). Wording differed, Holm p = .104
Interaction non-interference (1 to 7)	Control: 5.07 (0.80)*; HEAR: 6.00 (1.00). Wording differed, Holm p = .027

Note: TRS = Trade-off Reasoning Score, RES = Reflective Engagement Score, and PFUC = Perceived Feedback Utility/Congruence. TRS used exploratory single-coder ratings. RES and PFUC describe the HEAR group only. Asterisks mark control items that were conceptually similar but not identically worded. The corresponding differences are supplementary trends.

C. Item-Level HETU Results

As a supplementary analysis, none of the six HETU items showed a between-condition difference after Holm adjustment. The largest raw difference concerned the item about moving beyond a binary choice, where HEAR changed less than control. Its unadjusted p value was .021, and its Holm-adjusted p value was .127. No item remained significant after correction. The intervals were wide because the cells were small and starting scores differed. Figure 5 presents the item-level estimates.

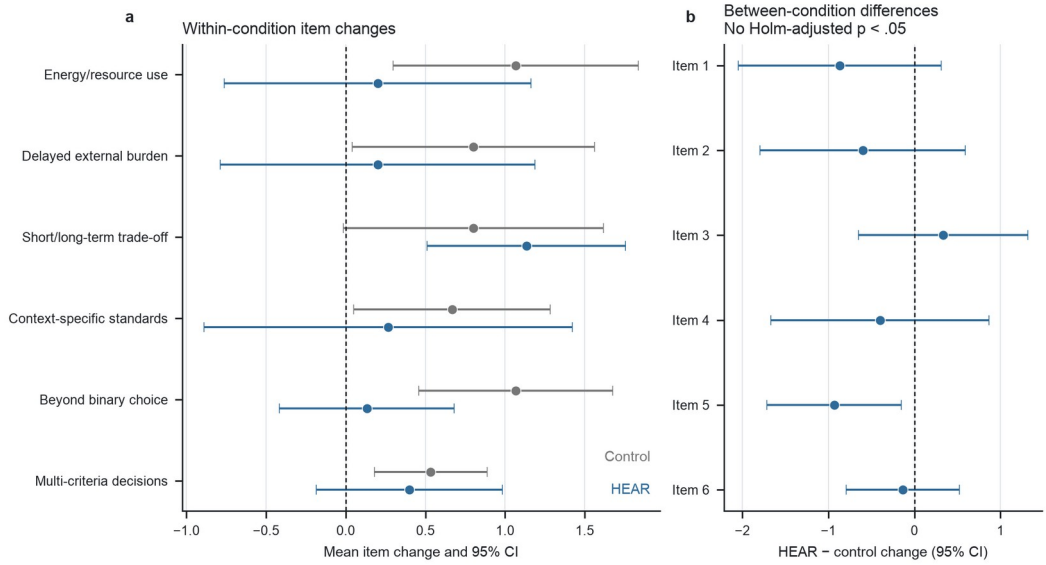


Fig. 5. HETU item changes and between-condition differences

D. Mechanism Outcomes and Post-Test Experience

The shared open response produced a higher exploratory Trade-off Reasoning Score in HEAR than in control. HEAR scores were M = 1.20, SD = 0.94, median = 1.0, and IQR = 0.5 to 2.0. Control scores were M = 0.07, SD = 0.26, median = 0, and IQR = 0 to 0. The Mann–Whitney result was U = 190.5, p < .001, with rank-biserial r = .69. A total of 6/15 HEAR responses (40%) scored at least 2, compared with 0/15 control responses (Fisher p = .017). HEAR participants were therefore more likely to articulate relations among collection preservation, energy use, and ecological consequences. The rubric was developed after data collection and applied once, so this result remains exploratory.

Within HEAR, RES averaged 5.77 (SD = 1.15, 95% bootstrap CI, 5.20 to 6.30, Spearman–Brown = .75). PFUC averaged 6.03 (SD = 0.93, 95% bootstrap CI, 5.57 to 6.47, Spearman–Brown = .82). A total of 11/15 participants had RES scores of at least 5, and 14/15 had PFUC scores of at least 5. Mean agency preservation was 5.93 (SD = 1.10), and mean non-interference was 6.00 (SD = 1.00). These measures test whether HEAR produced the intended bounded reflective interaction. They complement the learning outcome measured by HETU.

Figure 6 presents three concepts that could be compared approximately across the two post-tests. HEAR minus control differences were 0.53 for recognizing that no single answer existed (Holm p = .208), 0.80 for agency (Holm p = .104), and 0.93 for non-interference (Holm p = .027). Only non-interference met the adjusted threshold. Because the two questionnaires did not use identical wording, these values are not treated as formal between-condition effects. Their direction is consistent with the high agency and non-interference ratings within HEAR.

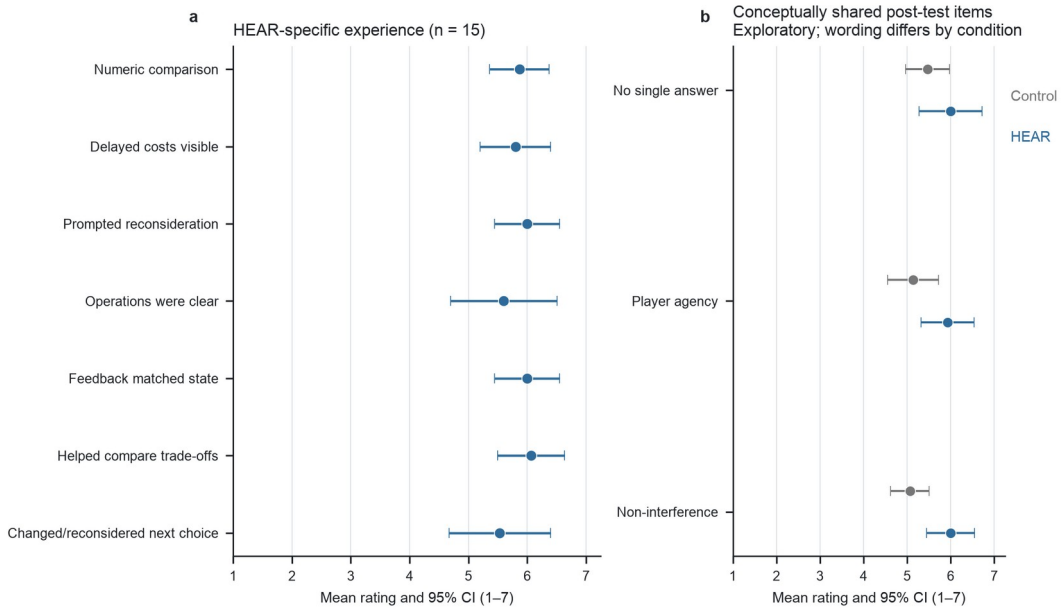


Fig. 6. HEAR mechanism experience and supplementary between-condition trends

IV. Discussion

This pilot produced two types of evidence that complemented each other but did not answer the same question. Within this sample and measurement window, HEAR did not produce a detectable additional short-term improvement in HETU after baseline adjustment. The mechanism measures showed that HEAR participants expressed more multi-factor trade-offs in the shared open-ended question. They also gave high ratings for reflective engagement, feedback-state congruence, and decision autonomy. Compared with the short-term change in HETU, the mechanism evidence for immediate reflection and trade-off expression was clearer. The mechanism analysis supports a narrower conclusion. HEAR participants described explicit trade-offs more often in their open responses and rated the feedback as closely related to the current game state. These results suggest that the feedback may have encouraged them to reconsider their choices without clearly reducing their decision autonomy. HEAR should therefore not be assessed only through short-term learning scores. The framework also provides an auditable form of reflective feedback while leaving the final decision to the player.

This pilot included only 30 participants. Allocation was not randomized, one coder scored the open responses, and some shared items used different wording across conditions. Future studies should randomize players after the pre-test, use the same wording for shared constructs such as autonomy and non-interference, and preregister the coding rubric for open responses. They should also add player interviews, use larger validation samples, and include behavioral tasks. Event logs should record model requests and players' later choices.

V. Conclusion

HEAR provides a state-constrained Large Language Model feedback framework for museum sustainability decisions. A deterministic engine controls states and scores, while the model can only select reflective prompts; every substantive choice remains with the player. In this pilot, HEAR did not produce a detectable additional short-term HETU gain over the control game. Exploratory evidence instead pointed to more explicit heritage-ecology trade-off reasoning, high feedback congruence, preserved agency, and immediate reconsideration. Because the sample was small, baseline balance was imperfect, and open responses were rated once, the findings support the mechanism rather than a broad learning claim. Larger studies should use harmonized measures and linked behavioral logs.

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